I. System Model

A mono-static MIMO-OFDM ISAC system, in which a dual-functional BS equipped with two separate UPAs of $N_{\rm tx}$ transmit antennas and $M_{\rm rx}$ receive antennas simultaneously performs downlink multi-user communications and radar target parameter estimation. Specifically, the BS transmits OFDM waveforms to communicate with K single-antenna users and simultaneously illuminates multiple point-like targets. Meanwhile, the received echo signals are processed to estimate the angle-range-velocity of targets.

A. Transmitted Signal Model

In the considered OFDM system, the carrier frequency is $f_{\rm c}$, and the wavelength is $\lambda_{\rm c} = c/f_{\rm c}$, where c denotes the speed of light. There are $N_{\rm s}$-subcarriers with frequency spacing $\Delta f = 1/T_{\rm d}$, where $T_{\rm d}$ is the OFDM symbol duration. For the l-th OFDM symbol, we denote the baseband signal transmitted on the i-th subcarrier as $\mathbf{x}_i[l] \in \mathbb{C}^{N_{\rm tx}}$, $i = 0, 1, \ldots, N_{\rm s} - 1$, $l = 0, 1, \ldots, L - 1$, where L is the frame length of one CPI.

The $N_{\rm s}$ $N_{\rm tx}$-dimensional frequency-domain baseband signals $\mathbf{x}_i[l]$ collected in different subcarriers are rearranged into $N_{\rm tx}$ $N_{\rm s}$-dimensional vectors collected from different transmit antennas. Then, $N_{\rm s}$-point inverse DFT (IDFT) processors are utilized to transform these frequency-domain vectors to the time domain. These signals are then arranged serially in chronological order, and an $N_{\rm cp}$-point CP of duration $T_{\rm cp}$ is inserted to avoid inter-symbol interference (ISI) for both communication and sensing. The CP length should be greater than the length of the channel impulse response to avoid ISI for downlink communications. In order to eliminate ISI at the sensing receiver, the CP duration should also be larger than the roundtrip delay between the BS and the furthest target. After conversion to analog, the baseband signal is expressed as

$$\widetilde{\mathbf{x}}(t) \triangleq \sum_{i=0}^{N_{\rm s}-1} \sum_{l=0}^{L-1} \mathbf{x}_i[l] e^{\jmath 2\pi i \Delta f t} \mathrm{rect}\left(\frac{t - lT}{T}\right), \tag{1}$$

where $T \triangleq T_{\rm d} + T_{\rm cp}$ is the total symbol duration and $\mathrm{rect}(t/T)$ denotes a rectangular pulse of duration T. Finally, the baseband analog signal is up-converted to the radio frequency (RF) domain via $N_{\rm tx}$ RF chains with carrier frequency $f_{\rm c}$ and then emitted through the antennas.

B. Communication Signal Model

After propagating through downlink communication channels, the OFDM signals are received by the single-antenna users and then demodulated into communication symbols. The communication receiver employs a series of operations including down-conversion, analog-to-digital converting (ADC), CP removal, serial-to-parallel conversion, and an $N_{\rm s}$-point DFT. For the k-th user, the frequency-domain signal on the i-th subcarrier during the l-th OFDM symbol is written as

$$y_{i,k}[l] \triangleq \mathbf{h}_{i,k}^H \mathbf{x}_i[l] + z_{i,k}[l], \tag{2}$$

where the vector $\mathbf{h}_{i,k} \in \mathbb{C}^{N_{\rm tx}}$ denotes the frequency domain channel between the BS and the k-th user, and $z_{i,k} \in \mathcal{CN}(0, \sigma_{\rm c}^2)$ denotes additive white Gaussian noise (AWGN).

C. Sensing Signal Model

From the radar sensing perspective, the dual-functional BS attempts to estimate the angle-range-velocity information of multiple point-like targets by processing the received echo signals. We assume that there are Q targets within the area of interest, and the azimuth/elevate angle-range-velocity information of the q-th target is denoted as θ_q, ϕ_q, R_q, and v_q, respectively, $q \in \mathcal{Q} \triangleq \{1, \ldots, Q\}$. Note that the angle of arrival (AoA) and the angle of departure (AoD) are both equal.

The emitted signals will first reach the Q targets and then be reflected back to the receive antennas of the BS. During this process, signals will experience the relative propagation delay between the transmit/receive antennas, the round-trip propagation delay between the array reference points, and potentially a Doppler frequency shift. Thus, the baseband echo signal received by the $(m_x,\, m_y)$-th antenna can be expressed as

$$\widetilde{y}(m_x, m_y, t) \triangleq \sum_{q=1}^{Q} \beta_q \mathbf{a}^H(\theta_q, \phi_q) \widetilde{\mathbf{x}}(t - 2R_q/c) e^{-\jmath \frac{2\pi}{\lambda_c} d(m_x \sin\theta_q \cos\phi_q + m_y \sin\theta_q \sin\phi_q)} e^{\jmath 2\pi f_{\mathrm{D},q} t} + \widetilde{z}(m_x, m_y, t), \quad (3)$$

where $\{m_x, m_y\} = 0, 1, \ldots, M_{\mathrm{rx}} - 1, M_{\mathrm{ry}} - 1$. In (3), β_q is the attenuation coefficient of the q-th target with power $\mathbb{E}\{|\beta_q|^2\} = \sigma_\beta^2$; $\mathbf{a}^H(\theta, \phi) \triangleq \mathbf{a}_x(\theta_q, \phi_q) \otimes \mathbf{a}_y(\theta_q, \phi_q)$ is the transmit steering vector for AoD θ, with

$$\mathbf{a}_x(\theta_q, \phi_q) = \begin{bmatrix} 1 \\ e^{-\jmath \frac{2\pi}{\lambda_c} d_{\mathrm{t},x} \sin\theta_q \cos\phi_q} \\ \vdots \\ e^{-\jmath \frac{2\pi}{\lambda_c} (N_x - 1) d_{\mathrm{t},x} \sin\theta_q \cos\phi_q} \end{bmatrix}, \quad (4)$$

$$\mathbf{a}_y(\theta_q, \phi_q) = \begin{bmatrix} 1 \\ e^{-\jmath \frac{2\pi}{\lambda_c} d_{\mathrm{t},y} \sin\theta_q \sin\phi_q} \\ \vdots \\ e^{-\jmath \frac{2\pi}{\lambda_c} (N_y - 1) d_{\mathrm{t},y} \sin\theta_q \sin\phi_q} \end{bmatrix}. \quad (5)$$

where d represents antenna spacing; $f_{\mathrm{D},q} = 2v_q f_{\mathrm{c}}/c$ indicates the Doppler frequency; and $\widetilde{z}(m, t)$ denotes independently and identically distributed (i.i.d.) AWGN. It is generally assumed that the attenuation coefficient and the angle-range-velocity of the targets are constant during one CPI. Considering that the signal bandwidth is usually much smaller than the carrier frequency, the phase shifts along the spatial axis and the Doppler phase shift within one OFDM symbol are respectively assumed to be identical on all subcarriers. In addition, only first-order reflections from the targets are considered due to high attenuation.

The baseband analog signals (3) are first processed by ADCs with sampling frequency $F_{\mathrm{s}} \triangleq N_{\mathrm{s}} \Delta f$, and the CP is removed from the digital samples. The resulting sampled echo signals for the l-th symbol-slot can be written as

$$\widehat{y}(m_x, m_y, j, l) \triangleq \widetilde{y}\left(m_x, m_y, lT + \frac{j}{F_{\mathrm{s}}} + T_{\mathrm{cp}} \right), \quad (6)$$

where $j = 0, \ldots, N_{\mathrm{s}} - 1$ is the sample index. After serial-to-parallel conversion and application of an N_{s}-point DFT along the sample index dimension, the echo signals are finally be obtained. The corresponding mathematical expression for the echo signal received by the (m_x, m_y)-th receive antenna on the i-th subcarrier is acquired by substituting expressions (1) and (3) into (6) and performing the DFT:

$$
\begin{aligned}
y(m_x, m_y, i, l) &\triangleq \sum_{q=1}^{Q} \beta_q \mathbf{a}^H(\theta_q, \phi_q)\mathbf{x}_i[l] e^{-\jmath \frac{2\pi}{\lambda_\mathrm{c}} d(m_x \sin\theta_q \cos\phi_q + m_y \sin\theta_q \sin\phi_q)} e^{-\jmath 4\pi i \Delta f R_q / c} e^{\jmath 4\pi l T v_q f_\mathrm{c}/c} + z(m_x, m_y, i, l) \\
&= \sum_{q=1}^{Q} \beta_q \mathbf{a}^H(\theta_q, \phi_q)\mathbf{x}_i[l] e^{\jmath m_x \omega_{\mathrm{a},x}(\theta_q,\phi_q)} e^{\jmath m_y \omega_{\mathrm{a},y}(\theta_q,\phi_q)} e^{\jmath i \omega_\mathrm{r}(R_q)} e^{\jmath l \omega_\mathrm{v}(v_q)} + z(m_x, m_y, i, l). \qquad (7a)
\end{aligned}
$$

where $z(m_x, m_y, i, l) \sim \mathcal{CN}(0, \sigma_\mathrm{s}^2)$ denotes the DFT of the AWGN. For conciseness, we respectively define the digital frequencies related to the angle, range, and velocity of the q-th target as

$$\omega_{\mathrm{a},x}(\theta_q, \phi_q) \triangleq -\frac{2\pi d}{\lambda_\mathrm{c}} \sin\theta_q \cos\phi_q, \tag{8a}$$

$$\omega_{\mathrm{a},y}(\theta_q, \phi_q) \triangleq -\frac{2\pi d}{\lambda_\mathrm{c}} \sin\theta_q \sin\phi_q, \tag{8b}$$

$$\omega_\mathrm{r}(R_q) \triangleq -4\pi \Delta f R_q / c, \tag{8c}$$

$$\omega_\mathrm{v}(v_q) \triangleq 4\pi T v_q f_\mathrm{c} / c. \tag{8d}$$

In this paper, we focus on the radar sensing problem of estimating the angle-range-velocity parameters of the targets based on the received echo signals in (7) within one CPI. We see that the target angle, range, and velocity are determined by the complex sinusoids of digital frequencies $\omega_{\mathrm{a,x}}(\theta_q, \phi_q)$, $\omega_{\mathrm{a,y}}(\theta_q, \phi_q)$, $\omega_\mathrm{r}(R_q)$, and $\omega_\mathrm{v}(v_q)$, respectively. However, these sinusoidal functions are multiplied by the signal-dependent coefficient $\mathbf{a}^H(\theta_q, \phi_q)\mathbf{x}_i[l]$, which changes with the transmitted signals and the target AoDs to be estimated. Thus, the target parameters cannot be directly extracted by performing spectral analysis on (7). In order to tackle this difficulty and improve the parameter estimation performance, we propose a novel joint angle-range-velocity estimation method to remove the influence of the signal-dependent coefficients and fully exploit the received echo signals during one CPI.

II. Joint Angle-Range-Velocity Estimation

In this section, we focus on extracting the angle-range-velocity information of potential targets from the echo signals (7). A novel joint estimation method is proposed to fully exploit the 3D data cube during one CPI and jointly estimate the target parameters. Following traditional radar terminology, we refer to the subcarrier dimension, i.e., $\{i = 0, 1, \ldots, N_\mathrm{s} - 1\}$, as fast-time and the symbol slot dimension, i.e., $\{l = 0, 1, \ldots, L - 1\}$, as slow-time. The dimension corresponding to the receive antennas, i.e., $\{\{m_x, m_y\} = 0, 1, \ldots, M_\mathrm{rx} - 1, M_\mathrm{ry} - 1\}$, is referred to as the spatial dimension.

A. Echo Signal Observations

Based on the expression for the echo signals in (7), we have the following observations.

- Along the spatial dimension for any given i, l, the echo signals can be regarded as the summation of Q complex sinusoids with frequencies $\omega_{\mathrm{a},x}(\theta_q, \phi_q)$, $\omega_{\mathrm{a,y}}(\theta_q, \phi_q)$ and constant amplitudes. Thus, the frequencies along the spatial dimension are determined by only the target angle.

- Along the fast-time dimension for any given m, l, the echo signals are composed of Q sinusoids with frequencies $\omega_{\rm r}(R_q)$, multiplied by the signal-dependent coefficients $\mathbf{a}^H(\theta_q,\phi_q)\mathbf{x}_i[l]$. Thus, the digital frequencies along the fast-time dimension are determined by the target range and signal-dependent coefficients.
- Along the slow-time dimension for any given m, i, the echo signals also consist of Q sinusoidal functions of frequencies $\omega_{\rm v}(v_q)$, also multiplied by the signal-dependent coefficients $\mathbf{a}^H(\theta_q,\phi_q)\mathbf{x}_i[l]$. As a result, the digital frequencies along the slow-time dimension are determined by the target velocity and signal-dependent coefficients.
- The signal-dependent coefficient $\mathbf{a}^H(\theta_q,\phi_q)\mathbf{x}_i[l]$ is related to the angle θ_q to be estimated and the transmitted signal $\mathbf{x}_i[l]$. In the considered ISAC system, the transmitted dual-functional signal $\mathbf{x}_i[l]$ is embedded with random communication symbols, which means that $\mathbf{x}_i[l]$ will change randomly for different indices i and l. These signal-dependent coefficient will cause random fluctuations in the fast- and slow-time dimensions, and thus will significantly hinder the analysis needed to obtain the range and velocity information.

Based on the above observations, it is clear that the angle-range-velocity information is fused across different dimensions of the data cube, and the signal-dependent coefficients $\mathbf{a}^H(\theta_q,\phi_q)\mathbf{x}_i[l]$ are the major obstacle to extracting the target parameters. In the following, a novel joint estimation method is developed to extract the angle-range-velocity information coupled with the troublesome signal-dependent coefficient.

B. Step 1: Spectral Analysis along the Spatial Dimension

We first analyze the frequency spectrum of the data cube along the spatial dimension to determine the target angles (θ_q,ϕ_q). In order to facilitate the analysis, the echo signal (7) is first re-arranged as

$$y(m_x,m_y,i,l)=\sum_{q=1}^{Q}\mathcal{A}(q,i,l)e^{\jmath m_x\omega_{{\rm a},x}(\theta_q,\phi_q)}e^{\jmath m_y\omega_{{\rm a},y}(\theta_q,\phi_q)}+z(m_x,m_y,i,l), \tag{9}$$

where $\mathcal{A}(q,i,l)$ does not depend on the spatial index m_x and m_y, and is defined as

$$\mathcal{A}(q,i,l)\triangleq\beta_q\mathbf{a}^H(\theta_q,\phi_q)\mathbf{x}_i[l]e^{\jmath i\omega_{\rm r}(R_q)}e^{\jmath l\omega_{\rm v}(v_q)}. \tag{10}$$

According to (9), each sequence $\{y(m_x,m_y,i,l),\ m_x=0,\ldots,M_{\rm rx}-1,\ m_y=0,\ldots,M_{\rm ry}-1\},\quad\forall i,\ \forall l$, along the spatial dimension can be viewed as a sum of noise and Q complex sinusoids with angle-dependent frequencies $\omega_{{\rm a},x}(\theta_q,\phi_q)$ and $\omega_{{\rm a},y}(\theta_q,\phi_q)$, and amplitude $\mathcal{A}(q,i,l)$. While one can easily obtain the angles of potential targets via spectral analysis by using only one such sequence, this ignores information from other available echo signals. Furthermore, although the estimated angles could be used to remove the signal-dependent coefficients, the angle estimation error will be propagated and even amplified in the estimation of the other parameters. Therefore, instead of directly obtaining angle information, the main purpose of spectrum analysis along the spatial dimension is to capture the characteristics of echo signals in different angular bins.

The angular spectral analysis can be conducted by various algorithms, such as the DFT, MUSIC, ESPRIT, compressed sensing based methods, etc. Each of these has its strengths and limitations. Due to its superior robustness and versatility, we will use the standard DFT approach to develop the proposed joint estimation method that fully utilizes all received echoes in one CPI. In particular, by applying an $N_{\mathrm{a},x}$-point and an $N_{\mathrm{a},y}$-point normalized DFT on $\{y(m_x, m_y, i, l), m_x = 0, \ldots, M_{\mathrm{rx}}-1,\ m_y = 0, \ldots, M_{\mathrm{ry}}-1\}$, where $N_{\mathrm{a},x} \geq M_{\mathrm{rx}}$ and $N_{\mathrm{a},y} \geq M_{\mathrm{ry}}$, we can obtain the sequences $\{Y_{i,l}(n_{\mathrm{a},x}, n_{\mathrm{a},y}),\ n_{\mathrm{a},x} = -\left\lfloor \frac{N_{\mathrm{a},x}}{2} \right\rfloor, \ldots, \left\lfloor \frac{N_{\mathrm{a},x}}{2} \right\rfloor - 1, n_{\mathrm{a},y} = -\left\lfloor \frac{N_{\mathrm{a},y}}{2} \right\rfloor, \ldots, \left\lfloor \frac{N_{\mathrm{a},y}}{2} \right\rfloor - 1,\ \forall i,\ \forall l\}$, given by

$$Y_{i,l}(n_{\mathrm{a},x}, n_{\mathrm{a},y}) = \frac{1}{M_{\mathrm{rx}} M_{\mathrm{ry}}} \sum_{m_x=0}^{M_{\mathrm{rx}}-1} \sum_{m_y=0}^{M_{\mathrm{ry}}-1} y(m_x, m_y, i, l) e^{-\jmath m_x \widetilde{\omega}_{\mathrm{a},x}(n_{\mathrm{a},x})} e^{-\jmath m_y \widetilde{\omega}_{\mathrm{a},y}(n_{\mathrm{a},y})},, \tag{11}$$

where the $(n_{\mathrm{a},x}, n_{\mathrm{a},y})$-th frequency component ranging from $-\pi$ to π is defined as

$$\widetilde{\omega}_{\mathrm{a},x}(n_{\mathrm{a},x}) \triangleq \frac{2\pi n_{\mathrm{a},x}}{N_{\mathrm{a},x}}, \tag{12}$$

$$\widetilde{\omega}_{\mathrm{a},y}(n_{\mathrm{a},y}) \triangleq \frac{2\pi n_{\mathrm{a},y}}{N_{\mathrm{a},y}}. \tag{13}$$

Using the spectral analysis in (11), each echo signal (9), composed of Q complex sinusoids with frequency $\omega_{\mathrm{a},x}(\theta_q, \phi_q)$ and $\omega_{\mathrm{a},y}(\theta_q, \phi_q)$ and amplitude $\mathcal{A}(q, i, l)$, is converted into the summation of $N_{\mathrm{a},x} N_{\mathrm{a},y}$ complex sinusoids with frequency $\widetilde{\omega}_{\mathrm{a},x}(n_{\mathrm{a},x})$ and $\widetilde{\omega}_{\mathrm{a},y}(n_{\mathrm{a},y})$ and amplitude $Y_{i,l}(n_{\mathrm{a}})$ as

$$y(m_x, m_y, i, l) = \sum_{n_{\mathrm{a},x}=-\lfloor N_{\mathrm{a},x}/2 \rfloor}^{\lfloor N_{\mathrm{a},x}/2 \rfloor - 1} \sum_{n_{\mathrm{a},y}=-\lfloor N_{\mathrm{a},y}/2 \rfloor}^{\lfloor N_{\mathrm{a},y}/2 \rfloor - 1} Y_{i,l}(n_{\mathrm{a},x}, n_{\mathrm{a},y}) e^{\jmath m_x \widetilde{\omega}_{\mathrm{a},x}(n_{\mathrm{a},x})} e^{\jmath m_y \widetilde{\omega}_{\mathrm{a},y}(n_{\mathrm{a},y})}.. \tag{14}$$

Thus, the echo signals are extracted into different bins corresponding to the angles of potential targets Specifically, the power present in the $(n_{\mathrm{a},x}, n_{\mathrm{a},y})$ angular bin is $\left|Y_{i,l}(n_{\mathrm{a},x}, n_{\mathrm{a},y})\right|^2$, which is related to the probability of the existence of targets whose angle-dependent frequencies $\omega_{\mathrm{a},x}(\theta_q, \phi_q)$ and $\omega_{\mathrm{a},y}(\theta_q, \phi_q)$ are approximately equal to $\widetilde{\omega}_{\mathrm{a},x}(n_{\mathrm{a},x})$ and $\widetilde{\omega}_{\mathrm{a},y}(n_{\mathrm{a},y})$. Next, we will utilize this characteristic to facilitate removal of the signal-dependent coefficient.

For the angles within bin $(n_{\mathrm{a},x}, n_{\mathrm{a},y})$, we have the following approximation

$$\omega_{\mathrm{a},x}(\theta_q, \phi_q) \approx \widetilde{\omega}_{\mathrm{a},x}(n_{\mathrm{a},x}), \quad \forall q \in \mathcal{Q}_{n_{\mathrm{a},x}, n_{\mathrm{a},y}}, \tag{15}$$

$$\omega_{\mathrm{a},y}(\theta_q, \phi_q) \approx \widetilde{\omega}_{\mathrm{a},y}(n_{\mathrm{a},y}), \quad \forall q \in \mathcal{Q}_{n_{\mathrm{a},x}, n_{\mathrm{a},y}}. \tag{16}$$

The angles of the targets in the $(n_{\mathrm{a},x}, n_{\mathrm{a},y})$-th angular bin can be further approximated as

$$\sin\theta_q \cos\phi_q \approx -\frac{n_{\mathrm{a},x} \lambda_{\mathrm{c}}}{d N_{\mathrm{a},x}}, \quad \forall q \in \mathcal{Q}_{n_{\mathrm{a},x}, n_{\mathrm{a},y}}, \tag{17}$$

$$\sin\theta_q \sin\phi_q \approx -\frac{n_{\mathrm{a},y} \lambda_{\mathrm{c}}}{d N_{\mathrm{a},y}}, \quad \forall q \in \mathcal{Q}_{n_{\mathrm{a},x}, n_{\mathrm{a},y}}. \tag{18}$$

Accordingly, we define the representative direction cosines of the $(n_{\mathrm{a},x}, n_{\mathrm{a},y})$-th angular bin as

$$u_{n_{\mathrm{a},x}} \triangleq -\frac{n_{\mathrm{a},x} \lambda_{\mathrm{c}}}{d N_{\mathrm{a},x}}, \tag{19a}$$

$$v_{n_{\mathrm{a},y}} \triangleq -\frac{n_{\mathrm{a},y} \lambda_{\mathrm{c}}}{d N_{\mathrm{a},y}}. \tag{19b}$$

Substituting (9) and (10) into (11) and using the approximations in (16) and (17), the amplitude of the $(n_{\mathrm{a},x}, n_{\mathrm{a},y})$-th angular bin can be approximated as

$$\begin{aligned} Y_{i,l}(n_{\mathrm{a},x}, n_{\mathrm{a},y}) &\approx \sum_{q \in \mathcal{Q}_{n_{\mathrm{a},x}, n_{\mathrm{a},y}}} \mathcal{A}(q, i, l) \\ &\approx \sum_{q \in \mathcal{Q}_{n_{\mathrm{a},x}, n_{\mathrm{a},y}}} \beta_q \mathbf{a}^H(u_{n_{\mathrm{a},x}}, v_{n_{\mathrm{a},y}}) \mathbf{x}_i[l] e^{\jmath i \omega_{\mathrm{r}}(R_q)} e^{\jmath l \omega_{\mathrm{v}}(v_q)}, \end{aligned} \tag{20}$$

where the noise component is ignored to focus on extracting the desired parameters from the target echo signals. We observe that $Y_{i,l}(n_{\mathrm{a},x}, n_{\mathrm{a},y})$ consists of $Q_{n_{\mathrm{a},x},n_{\mathrm{a},y}}$ components, each of which is the signal-dependent coefficient $\mathbf{a}^H(u_{n_{\mathrm{a},x}}, v_{n_{\mathrm{a},y}})\mathbf{x}_i[l]$ multiplied by complex sinusoids whose frequencies are related to the range and velocity of the targets. Since the signal-dependent coefficient changes along both the fast- and slow-time dimensions, it hinders the estimation of range and velocity using standard spectral analysis algorithms. Thus, the next step is to eliminate the impact of this signal-dependent coefficient.

C. Step 2: Signal-Dependent Coefficients Removing

After spectral analysis along the spatial dimension, the amplitude of the echo signals with different angle-dependent frequencies is extracted. Then the signal-dependent term $\mathbf{a}^H(u_{n_{\mathrm{a},x}}, v_{n_{\mathrm{a},y}})\mathbf{x}_i[l]$ is removed to eliminate its influence on the spectral analysis along the fast- and slow-time dimensions to extract the estimates of range and velocity. Since the transmitted signals $\mathbf{x}_i[l]$ are known at the dual-functional BS, it would be straightforward to use this information together with the estimated target angle bins to remove the signal-dependent term $\mathbf{a}^H(u_{n_{\mathrm{a},x}}, v_{n_{\mathrm{a},y}})\mathbf{x}_i[l]$. However, only part of the 4D data cube is used in this sequential strategy, and in addition, the angle estimation errors will propagate and degrade the subsequent range and velocity estimation. Moreover, the method is sensitive to the ambiguity function associated with the transmitted signals, and its performance degrades due to the randomness of the communication symbols. To avoid these drawbacks, we propose to completely remove the signal-dependent term for each angular bin instead of only focusing on the few estimated target angule bins associated with potential targets. Thus, our approach performs a joint angle-range-velocity estimation to enhance performance by fully exploiting all echo signals during each CPI.

From equation (20), we observe that the signal-dependent term $\mathbf{a}^H(u_{n_{\mathrm{a},x}}, v_{n_{\mathrm{a},y}})\mathbf{x}_i[l]$ could be directly removed by point-wise division. However, since the magnitude of the signal-dependent term is not identical for different angular bins, directly dividing $Y_{i,l}(n_{\mathrm{a},x}, n_{\mathrm{a},y})$ by the signal-dependent coefficient will destroy the properties of the echo signals in the spatial dimension. In particular, the spatial characteristics of the echo signals are changed after dividing $Y_{i,l}(n_{\mathrm{a},x}, n_{\mathrm{a},y})$ by different amplitude values, which will deteriorate the angle estimation performance.

To eliminate the impact of the signal-dependent term, we modify the data to maintain the spatial characteristics of the echo signals, and not change the relationship between the powers of different angle bins. For this purpose, we introduce a scaling factor $\alpha_{n_{\mathrm{a},x},n_{\mathrm{a},y}}$ for the $(n_{\mathrm{a},x}, n_{\mathrm{a},y})$-th angular bin and propose to modify $Y_{i,l}(n_{\mathrm{a},x}, n_{\mathrm{a},y})$ as follows:

$$y_{i,l}(n_{\mathrm{a},x}, n_{\mathrm{a},y}) \triangleq \begin{cases} \dfrac{Y_{i,l}(n_{\mathrm{a},x}, n_{\mathrm{a},y})}{\alpha_{n_{\mathrm{a},x},n_{\mathrm{a},y}}\mathbf{a}^H(u_{n_{\mathrm{a},x}}, v_{n_{\mathrm{a},y}})\mathbf{x}_i[l]}, & \text{if } \mathbf{a}^H(u_{n_{\mathrm{a},x}}, v_{n_{\mathrm{a},y}})\mathbf{x}_i[l] \neq 0, \\ Y_{i,l}(n_{\mathrm{a},x}, n_{\mathrm{a},y}), & \text{if } \mathbf{a}^H(u_{n_{\mathrm{a},x}}, v_{n_{\mathrm{a},y}})\mathbf{x}_i[l] = 0, \end{cases} \tag{21}$$

where the selection is eliminate dividing by zero. Then, based on the above discussions, the scaling factor $\alpha_{n_{\mathrm{a},x},n_{\mathrm{a},y}}$ is chosen such that the power of the $(n_{\mathrm{a},x}, n_{\mathrm{a},y})$-th angular bin remains unchanged, i.e.,

$$\sum_{i,l} \left|y_{i,l}(n_{\mathrm{a},x}, n_{\mathrm{a},y})\right|^2 = \sum_{i,l} \left|Y_{i,l}(n_{\mathrm{a},x}, n_{\mathrm{a},y})\right|^2, \ \forall n_{\mathrm{a},x}, \ \forall n_{\mathrm{a},y}, \tag{22}$$

with leads to the following equation for the scaling factor:

$$\alpha_{n_{\mathrm{a},x},n_{\mathrm{a},y}} = \sqrt{\frac{\sum\limits_{i,l,\, \mathbf{a}^H(u_{n_{\mathrm{a},x}},v_{n_{\mathrm{a},y}})\mathbf{x}_i[l]\neq 0} \left|\frac{Y_{i,l}(n_{\mathrm{a},x},n_{\mathrm{a},y})}{\mathbf{a}^H(u_{n_{\mathrm{a},x}},v_{n_{\mathrm{a},y}})\mathbf{x}_i[l]}\right|^2}{\sum\limits_{i,l,\, \mathbf{a}^H(u_{n_{\mathrm{a},x}},v_{n_{\mathrm{a},y}})\mathbf{x}_i[l]\neq 0} |Y_{i,l}(n_{\mathrm{a},x},n_{\mathrm{a},y})|^2}}. \tag{23}$$

According to the approximation in (20), we can further write each sample in (21) as

$$y_{i,l}(n_{\mathrm{a},x},n_{\mathrm{a},y}) \approx \sum_{q\in\mathcal{Q}_{n_{\mathrm{a},x},n_{\mathrm{a},y}}} \beta_q/\alpha_{n_{\mathrm{a},x},n_{\mathrm{a},y}}\, e^{\jmath i\omega_{\mathrm{r}}(R_q)}\, e^{\jmath l\omega_{\mathrm{v}}(v_q)}, \tag{24}$$

where the special case with $\mathbf{a}^H(u_{n_{\mathrm{a},x}},v_{n_{\mathrm{a},y}})\mathbf{x}_i[l]=0$ can be ignored without influencing the result. Now we clearly see from equation (24) that the resulting sequences obtained along the fast-time dimension $\{y_{i,l}(n_{\mathrm{a},x},n_{\mathrm{a},y}),\ \forall i\}$ or the slow-time dimension $\{y_{i,l}(n_{\mathrm{a},x},n_{\mathrm{a},y}),\ \forall l\}$ are composed of $Q_{n_{\mathrm{a},x},n_{\mathrm{a},y}}$ sinusoidal functions whose frequencies are determined by the range R_q or the velocity v_q. Thus, it is straightforward to perform spectral analysis along the two dimensions and jointly estimate the desired angle-range-velocity information, as presented in the next subsection.

D. Step 3: Spectral Analysis along the Fast- and Slow-Time Dimensions and Joint Estimation

In this section, we show how to employ a 2D-DFT operation along the fast- and slow-time dimensions of the signal in (21) to extract the echo signals into different range and Doppler bins. Then, together with the previously obtained angular spectrum, a joint angle-range-velocity estimation from the three dimensions is proposed.

Specifically, a normalized $(N_{\mathrm{r}},N_{\mathrm{v}})$-point $(N_{\mathrm{r}}\geq N_{\mathrm{s}},\ N_{\mathrm{v}}\geq L)$ 2D-DFT is implemented on the obtained sequences $\{y_{i,l}(n_{\mathrm{a},x},n_{\mathrm{a},y}),\ \forall i,\ \forall l\}$,, yielding the sequences $\{Y(n_{\mathrm{a},x},n_{\mathrm{a},y},n_{\mathrm{r}},n_{\mathrm{v}}),\ n_{\mathrm{r}}=-N_{\mathrm{r}}+1,\ldots,0,\ n_{\mathrm{v}}=-\lfloor N_{\mathrm{v}}/2\rfloor,\ldots,\lfloor N_{\mathrm{v}}/2\rfloor-1\},\ \forall n_{\mathrm{a},x},\ \forall n_{\mathrm{a},y}$. The amplitude $Y(n_{\mathrm{a},x},n_{\mathrm{a},y},n_{\mathrm{r}},n_{\mathrm{v}})$ can be calculated as

$$Y(n_{\mathrm{a},x},n_{\mathrm{a},y},n_{\mathrm{r}},n_{\mathrm{v}}) = \frac{1}{N_{\mathrm{s}}L}\sum_{i=0}^{N_{\mathrm{s}}-1}\sum_{l=0}^{L-1} y_{i,l}(n_{\mathrm{a},x},n_{\mathrm{a},y})e^{-\jmath l\widetilde{\omega}_{\mathrm{v}}(n_{\mathrm{v}})}e^{-\jmath i\widetilde{\omega}_{\mathrm{r}}(n_{\mathrm{r}})}, \tag{25}$$

where the frequency components of the n_{r}-th range bin and the n_{v}-th Doppler bin are respectively defined as

$$\widetilde{\omega}_{\mathrm{r}}(n_{\mathrm{r}}) \triangleq \frac{2\pi n_{\mathrm{r}}}{N_{\mathrm{r}}}, \tag{26a}$$

$$\widetilde{\omega}_{\mathrm{v}}(n_{\mathrm{v}}) \triangleq \frac{2\pi n_{\mathrm{v}}}{N_{\mathrm{v}}}. \tag{26b}$$

After the steps described above, the echo signals are extracted into the $N_{\mathrm{a},x}\times N_{\mathrm{a},y}\times N_{\mathrm{r}}\times N_{\mathrm{v}}$ angular-range-Doppler bins with the amplitudes given. Therefore, we can jointly estimate the targets from these 4D bins via peak finding. In particular, we can apply cell-averaging constant false alarm rate (CA-CFAR) processing with a 4D cell to obtain several signal clusters with high amplitudes, and then choose the maximum value of

Algorithm 1 Proposed Joint Angle-Range-Velocity Estimator

Input: $y(m_x, m_y, i, l)$, N_{tx}, M_{rx}, N_{s}, L, N_{a}, N_{r}, N_{v}, d, λ_{c}, c, f_{c}, T, Δf, $\mathbf{x}_i[l]$, $\forall m_x, m_y, i, l$.

Output: θ_q, ϕ_q, R_q, v_q.

1: Perform $(N_{\mathrm{a},x}, N_{\mathrm{a},y})$-point 2D-DFT on $y(m_x, m_y, i, l)$ along the spatial dimension to obtain $Y_{i,l}(n_{\mathrm{a},x}, n_{\mathrm{a},y})$, $\forall i, l, n_{\mathrm{a},x}, n_{\mathrm{a},y}$ in (11).

2: Calculate scaling factor $\alpha_{n_{\mathrm{a},x}, n_{\mathrm{a},y}}$ in (23).

3: Remove signal-dependent coefficients to obtain $y_{i,l}(n_{\mathrm{a},x}, n_{\mathrm{a},y})$ in (21).

4: Perform 2D-DFT along the fast- and slow-time dimensions to obtain $Y(n_{\mathrm{a},x}, n_{\mathrm{a},y}, n_{\mathrm{r}}, n_{\mathrm{v}})$.

5: Find the peaks of $[Y(n_{\mathrm{a},x}, n_{\mathrm{a},y}, n_{\mathrm{r}}, n_{\mathrm{v}}), \ \forall n_{\mathrm{a},x}, n_{\mathrm{a},y}, n_{\mathrm{r}}, n_{\mathrm{v}}]$ to obtain the index set $\{\mathcal{T}_q, \forall q\}$.

6: Recover the estimated $\widetilde{\theta}_q$, $\widetilde{\phi}_q$, $\widetilde{R}_q$, and $\widetilde{v}_q$ by (27).

7: Return $\theta_q = \widetilde{\theta}_q$, $\phi_q = \widetilde{\phi}_q$, $R_q = \widetilde{R}_q$, and $v_q = \widetilde{v}_q$.

each cluster as the peak, whose 4D coordinate we denote by $\mathcal{T}_q$. Finally, the estimated angle-range-velocity of the q-th target, defined as $\widetilde{\theta}_q$, $\widetilde{\phi}_q$, $\widetilde{R}_q$, and $\widetilde{v}_q$, can be recovered using the 4D index

$$\widetilde{u}_q = -\frac{n_{\mathrm{a},x}\lambda_{\mathrm{c}}}{dN_{\mathrm{a},x}}, \tag{27a}$$

$$\widetilde{v}_q^{(\mathrm{dir})} = -\frac{n_{\mathrm{a},y}\lambda_{\mathrm{c}}}{dN_{\mathrm{a},y}}, \tag{27b}$$

$$\widetilde{\theta}_q = \arcsin\left(\sqrt{\widetilde{u}_q^2 + \left(\widetilde{v}_q^{(\mathrm{dir})}\right)^2}\right), \tag{27c}$$

$$\widetilde{\phi}_q = \mathrm{atan2}\left(\widetilde{v}_q^{(\mathrm{dir})}, \widetilde{u}_q\right), \tag{27d}$$

$$\widetilde{R}_q = -\frac{cn_{\mathrm{r}}}{2N_{\mathrm{r}}\Delta f}, \tag{27e}$$

$$\widetilde{v}_q = \frac{cn_{\mathrm{v}}}{2N_{\mathrm{v}}Tf_{\mathrm{c}}}. \tag{27f}$$

We emphasize that the angle-range-velocity information is estimated jointly in the 3D space, rather than estimating these quantities sequentially as in previous approaches, and thus the proposed algorithm achieves better estimation performance, especially when the targets are closely spaced along a particular dimension.